

University of Central Punjab -FOIT**Fall 2025 – C13****Quiz 4**
= 2**Total Marks = 10****CLO**

1. You have an integer counter initialized to 128. Using inline assembly, divide counter by 4 using a shift-right instruction and store the result back in counter. (5)

```
#include <iostream>

using namespace std;

int main() {

    int counter = 128;

    __asm {

        mov eax, counter ; move counter into eax

        shr eax, 2      ; shift right by 2 bits (divide by 4)

        mov counter, eax ; store result back into counter

    } cout << "Result: " << counter << endl; // should print 32

    return 0; }
```

2. Using inline assembly, add 7 to every element of the 2×2 integer array matrix and display the results. (5)

```
#include <iostream>
using namespace std;
int main() {
    int matrix[2][2] = { {1, 2}, {3, 4} };
    __asm {
        lea esi, matrix    // Load address of matrix into ESI
        mov ecx, 4          // Total number of elements (2x2 = 4)
    loop_start:
        mov eax, [esi]      // Load current element
        add eax, 7          // Add 7
        mov [esi], eax      // Store back
        add esi, 4          // Move to next element (sizeof(int) = 4)
        loop loop_start    // Repeat ECX times
    } for (int i = 0; i < 2; i++) {
        for (int j = 0; j < 2; j++) {
```

```
        cout << matrix[i][j] << " ";  
    }    cout << endl;  
}    return 0; }
```

Quiz 5**Total Marks = 10****CLO****= 3**

1. Two MMX registers contain: (2)

mm0 = 0x9000 0xF000 0x4000 0x5000

mm1 = 0x9000 0x2000 0x6000 0x4000

What will be the result of paddw mm0, mm1?

0x2000 0x1000 0xA000 0x9000

2. Two MMX registers contain: (2)

mm0 = 0x2000, 0x8000, 0x4000, 0x7000

mm1 = 0x1000, 0x2000, 0x8000, 0x9000

What will be the result of pmulhw mm0, mm1?

0x2000 0xF000 0xE000 0xCE00

3. Two MMX registers contain: (2)

mm0 = 0xF0F0, 0xAAAA, 0x1234, 0xFFFF

mm1 = 0x0F0F, 0x5555, 0x00FF, 0x0F0F

What will be the result of por mm0, mm1?

0xFFFF 0xFFFF 0x12FF 0xFFFF

4. Two MMX registers contain: (4)

mm0 = 0x0102030405060708

mm1 = 0x0807060504030201

What will be the result of the following sequence of instructions?

```
__asm{  
    paddw mm0, mm1  
    psrlb mm0, 2  
}
```

Name: _____

Reg #: _____

Paddb: 0x0909 0x0909 0x0909 0x0909

Psrlb: 0x0202 0x0202 0x0202 0x0202